PROCEEDINGS OF SCIENCE

How LHCb is shedding light on the existence of dark sector portals

Lisa Fantini[*,a] **on behalf of the LHCb collaboration**

[a] *INFN Perugia,*
Via A. Pascoli, Perugia, Italy

E-mail: lisa.fantini@cern.ch

Although suggested by cosmological and astrophysical observables, no dark matter candidate has been observed to date. Potential mediators between the visible and dark sectors are receiving more and more attention since they offer the opportunity of discovering the nature of dark matter at collider experiments. The LHCb experiment, originally built for $b-$ and $c-$physics, has proven to be uniquely sensitive to low-mass resonances. A focus is set on the search for Dark Photons in leptonic decays, hidden-sector bosons, and the newly published search for axion-like-particles. The prospects for those searches with the large Run 3 dataset will be shown as well.

The European Physical Society Conference on High Energy Physics (EPS-HEP2025)
7-11 July 2025
Marseille, France

© Copyright owned by the author(s) under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License (CC BY-NC-ND 4.0) All rights for text and data mining, AI training, and similar technologies for commercial purposes, are reserved. ISSN 1824-8039 . Published by SISSA Medialab.

PoS(EPS-HEP2025)043

1. Introduction

The Standard Model (SM) of particle physics is a very successful theoretical framework, however it leaves a number of fundamental questions unanswered, among which is the nature of dark matter (DM). DM constitutes about 27% of the energy density of the Universe according to the cosmological observations of the cosmic microwave background, the galaxy rotation curves, and the gravitational lensing [1]. While gravitational effects provide robust evidence for DM, no particle candidate within the SM can account for it.

This motivates theoretical extensions of the SM that introduce new sectors containing particles neutral under ordinary gauge interactions. These "dark sectors" can contain a variety of states that connect the dark and visible worlds. A common feature of such mediators is that their couplings to SM particles are extremely weak, which can also result in long lifetimes. This explains why dark matter has eluded direct detection despite decades of experimental effort.

Theoretical consistency strongly constrains the ways in which such sectors can couple to the SM. Gauge invariance and Lorentz symmetry imply that only four renormalizable "portals" exist: the vector, Higgs, axion and neutrino portals. Each of these portals has distinctive phenomenology and can be probed at colliders and fixed-target experiments, as well as through cosmological and astrophysical observations [2, 3].

The LHCb experiment [4] provides unique sensitivity to dark-sector portals. Its forward geometry ($2 < \eta < 5$), excellent vertex and momentum resolution, and low-momentum trigger thresholds make it especially well suited for low-mass searches and long-lived particle signatures. Moreover, with the full software trigger deployed in Run 3, LHCb can select even softer states and more displaced tracks, further extending its reach to unexplored regions of parameter space.

Dark-sector mediators can be produced in interactions of SM particles and decay into visible final states. This proceeding reviews LHCb's searches for such particles through the vector, scalar, and axion portals, presenting results from Run 1 (2010- 2012) and Run 2 (2015-2018), as well as prospects for Run 3, ongoing since 2022 with the upgraded detector.

2. Dark photon searches

Among the possible realizations of dark-sector mediators, the vector portal predicts the existence of a massive $U(1)$ gauge boson, the dark photon (A') [5]. Through kinetic mixing with the SM photon, this state acquires a small effective coupling to charged particles, enabling its production in proton–proton (pp) collisions and its decay into visible leptonic final states. The phenomenology of dark photon is governed by two parameters, its mass $m(A')$ and the kinetic-mixing strength, ε, which controls both the production rate and the decay width.

LHCb has performed a search for $A' \to \mu^+\mu^-$ using 5.5 fb^{-1} of data collected between 2016 and 2018 in pp collisions at a centre-of-mass energy of 13 TeV [6]. Two complementary regions were defined. In the prompt category, both muon tracks were required to originate from the pp primary vertex, providing sensitivity to short-lived A' decays; in this category, dark-photon masses between the dimuon threshold and 70 GeV were tested. In the displaced category, candidates were selected with an A' decay vertex significantly displaced from the pp primary vertex, probing long-lived dark photons with masses in the range $214 < m(A') < 350$ MeV/c^2 and lifetimes up to approximately 1 ps.

PoS(EPS-HEP2025)043

The analysis exploited the fact that prompt dark photons produced via γ–A' mixing follow the same kinematic distributions as off-shell photon production (γ^*), enabling a fully data-driven treatment of the SM background. In addition, the $\gamma^* \to \mu^+\mu^-$ process is used as the normalization channel, which results in the cancellation of many systematic effects. A scan of the dimuon invariant-mass spectrum, performed in steps of half the dimuon invariant mass resolution, was used to search for narrow resonances over a smooth background. The main background contributions for the prompt category are the irreducible γ^* decay component and the semileptonic decays of heavy-flavour hadrons. Moreover, events consistent with known dimuon resonances are removed by applying mass vetoes around their nominal masses. For displaced decays the main backgrounds are due to photon conversion in the detector material and double-semimuonic decays of heavy-flavour hadrons.

The resulting exclusion limits are presented in Figure 1, where the LHCb results are compared with previous fixed-target and collider searches. Exclusion limits are set at 95% confidence level (CL) on the kinetic-mixing parameter ε as a function of $m(A')$. The analysis achieved the most stringent limits to date in the ranges $214 < m(A') \lesssim 740$ MeV and $10.6 < m(A') \lesssim 30$ GeV. The displaced analysis has provided the first collider-based constraints on long-lived dark photons, independent of dedicated beam dump setups.

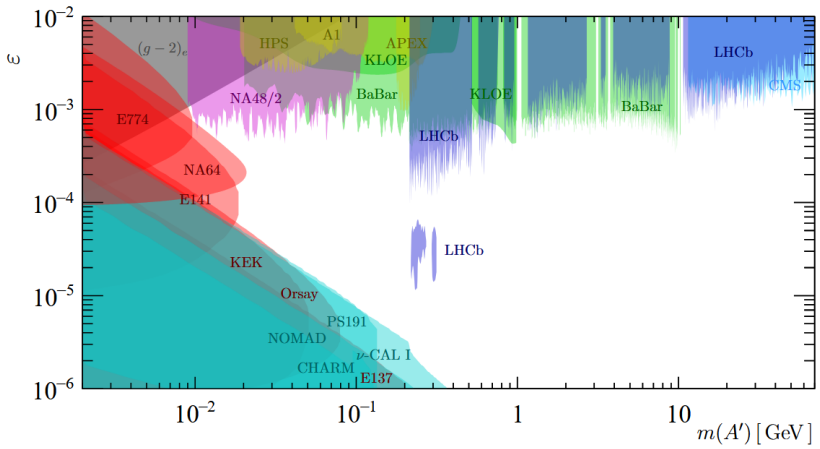


Figure 1: LHCb 95% CL exclusion limits on dark photons in the $(m_{A'}, \varepsilon)$ plane of the Minimal dark photon model, compared with results from other experiments. Figure from Ref. [6].

Future prospects are shown in Figure 2, which illustrates the expected sensitivity from the LHCb dark-photon projections [7]. The full software-trigger system implemented in Run 3 lowers lepton-momentum thresholds and enables inclusive $A' \to e^+e^-$ searches below the dimuon threshold, thus opening an unexplored mass window in the 10–200 MeV region. This region of parameter space overlaps with theoretically motivated scenarios where the kinetic-mixing parameter is loop-suppressed. The higher luminosity expected from Run 3 onwards will further improve sensitivity to both prompt and displaced A' decays.

PoS(EPS-HEP2025)043

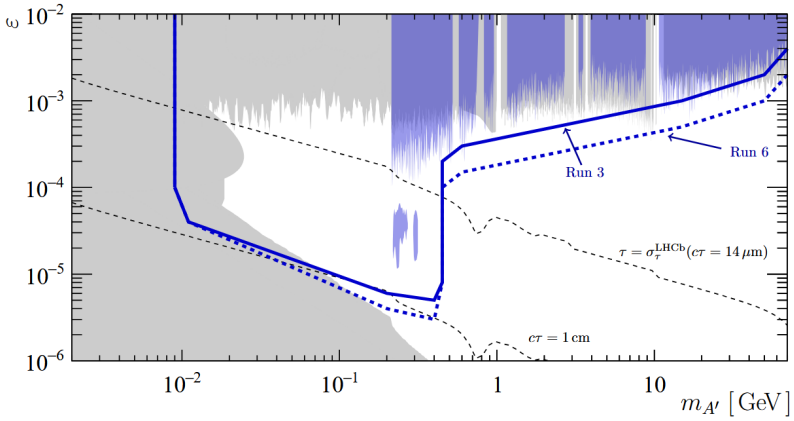


Figure 2: Projected LHCb sensitivity to dark photons in the $(m_{A'}, \varepsilon)$ plane for Run 3 (solid) and Run 6 (dashed). Existing experimental exclusions are shown in grey; dashed contours indicate representative lifetimes. Figure from Ref. [7].

3. Dark scalars in $b \to s\mu^+\mu^-$ transitions

The scalar portal connects the SM and hidden sectors via mixing between the Higgs field and a new scalar singlet χ. Depending on the mixing angle (θ), such a light scalar couples to quarks and leptons proportionally to their masses, and can therefore be produced in flavour-changing $b \to s$ transitions [8].

The LHCb collaboration has searched for hidden-sector bosons produced in the decays $B^0 \to K^{*0}(\to K^+\pi^-)\chi(\to \mu^+\mu^-)$ and $B^+ \to K^+\chi(\to \mu^+\mu^-)$, using 3 fb^{-1} of Run 1 data collected at centre-of-mass energies of 7 TeV and 8 TeV in pp collisions [9, 10].

The analysis of the neutral channel scans the dimuon invariant-mass spectrum in the range $214 < m(\chi) < 4350$ MeV/c^2. Events consistent with known resonances such as the ω, ϕ, J/ψ, and $\psi(2S)$ were vetoed, and the residual background was mainly due to the non-resonant $B^0 \to K^{*0}\mu^+\mu^-$ continuum. Separate selections were optimized for short- and long-lived candidates, exploiting the precise vertexing capabilities of the LHCb detector to identify displaced dimuon vertices up to lifetimes of about 100 ps. No significant excess was found, and limits were set on $\mathcal{B}(B^0 \to K^{*0}\chi) \times \mathcal{B}(\chi \to \mu^+\mu^-)$, reaching sensitivities down to $O(10^{-9})$ for promptly decaying bosons. The complementary search in the charged channel takes advantage of a decay topology free from the K^{*0} reconstruction ambiguity. This provides improved mass resolution and higher efficiency for displaced vertices. The analysis covered the same dimuon mass range and lifetimes. Backgrounds mainly originate from misidentified hadrons and semileptonic B decays producing genuine muons; these were suppressed using muon identification and vertex-quality criteria. The invariant-mass spectrum was scanned in fine steps of roughly half the detector resolution, and no localized excess was observed. The obtained limits are comparable to those obtained in the neutral mode, confirming the absence of signals over the full accessible mass range.

Figure 3 shows the exclusion limits from both analyses, together with cosmological and theoretical constraints. These results represent the most stringent constraints to date on the mixing angle θ in the mass range $214 < m_\chi < 3500$ MeV/c^2, excluding values above $\sim 10^{-4}$ in most of the explored region.

PoS(EPS-HEP2025)043

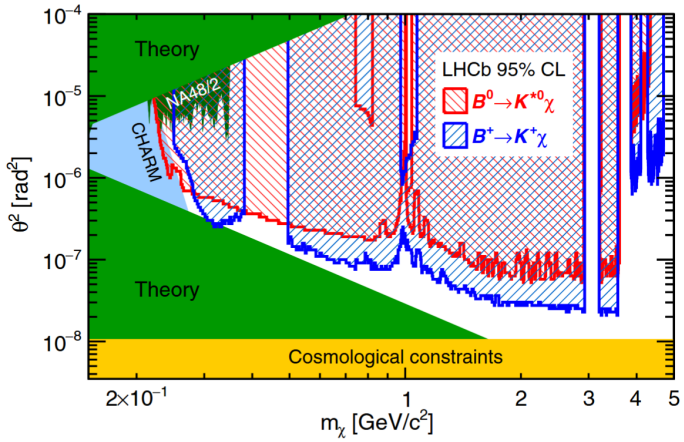


Figure 3: Exclusion limits at 95% CL from the LHCb searches for hidden-sector bosons in $B^0 \to K^0\chi$ (red) and $B^+ \to K^+\chi$ (blue) decays, shown in the (m_χ, θ^2) plane. The results are compared with previous experimental constraints and regions disfavoured by theoretical and cosmological considerations Figure from Ref. [10].

Future improvements are expected with Run 3 data. Figure 4 illustrates the projected sensitivity of the upgraded LHCb detector and the fully software-based trigger. These improvements will provide a significant gain in sensitivity, even when using only the LHCb standard "long-track" category, reconstructed from hits in the tracking detectors near the interaction point and in the downstream stations. In addition, including the contribution of the so-called downstream tracks, reconstructed without the upstream vertex-detector information, will further increase acceptance for displaced vertices [11].

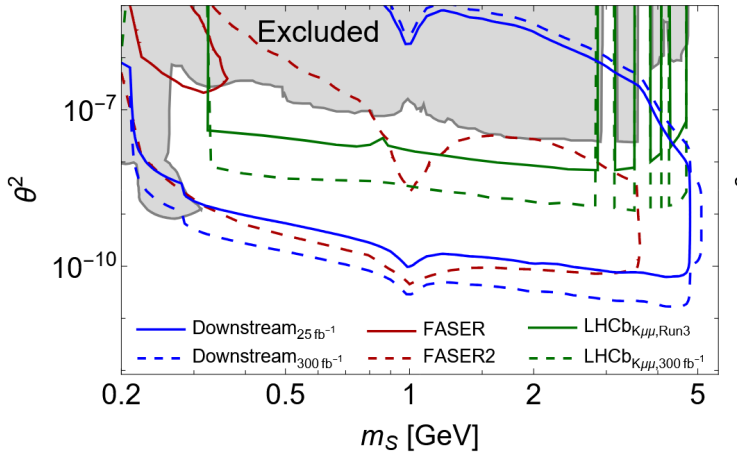


Figure 4: Projected sensitivities to long-lived scalar bosons in the (m_S, θ^2) plane ($m_S \equiv m_\chi$). LHCb projections for Run 3 (25 fb^{-1}) and for 300 fb^{-1} are shown in green for long tracks only and in blue with addition of downstream tracks, moreover are compared with expected sensitivity from FASER and FASER2 experiments (red). The grey area indicates the parameter space already excluded. Figure from Ref. [11]

PoS(EPS-HEP2025)043

4. Axion-like particles

Axion-like particles (ALPs) are pseudoscalar bosons that can act as mediators between the SM and a dark sector [12]. They naturally arise in many extensions of the SM that involve new spontaneously broken global symmetries. Unlike the QCD axion, introduced to solve the strong-CP problem, their mass and couplings are not fixed by the QCD scale, allowing for a wide range of phenomenologically viable scenarios. ALPs can couple to SM gauge bosons via this Lagrangian:

$$\mathcal{L} \supset \frac{a}{4\pi f_a}\left[\alpha_3 c_3\, G\tilde{G} + \alpha_2 c_2\, W\tilde{W} + \alpha_1 c_1\, B\tilde{B}\right] + \tfrac{1}{2} m_a^2 a^2 .$$

where, f_a is the symmetry-breaking scale controlling the overall coupling strength, while the terms in brackets describe the interactions of the ALP field a with gluons ($G\tilde{G}$) and electroweak gauge bosons ($W\tilde{W}$, $B\tilde{B}$). These interactions can lead to observable collider signatures, such as narrow diphoton or dilepton resonances.

A very recent LHCb search explores the decay $a \to \gamma\gamma$ using 2.1 fb^{-1} of pp collision data collected in 2018 at $\sqrt{s}$ = 13 TeV [13]. This analysis targets light pseudoscalars produced via gluon–gluon fusion and decaying into two photons, constituting the first fully neutral final-state search for ALPs in a hadronic environment. Signal candidates are built from electromagnetic calorimeter clusters consistent with prompt photon showers, while dedicated trigger lines select events with high-quality photon pairs. The search covers the diphoton invariant-mass range $4.9 < m_{\gamma\gamma} < 19.4$ GeV by scanning for narrow peaks above a smooth background. The dominant backgrounds originate from combinatorial photon pairs, mainly from $\pi^0 \to \gamma\gamma$ and other hadronic decays, as well as merged π^0 showers at high transverse energy. A multivariate classifier based on isolation variables is used to suppress these contributions, requiring photon candidates to be isolated from both charged tracks and nearby neutral energy deposits.

No significant excess is observed, and model-independent 95% CL upper limits are set on $\sigma(pp \to aX) \times \mathcal{B}(a \to \gamma\gamma)$, as shown in Figure 5 (left). The limits represent the first ALP constraints from a pp collider in the region below 20 GeV, improving upon previous LHCb measurements and providing complementary coverage to ATLAS and CMS searches. In Figure 5 (right) the results are interpreted in terms of the effective ALP–photon coupling ($g_{a\gamma\gamma}$) and the symmetry-breaking scale f_a, yielding the best constraint in the $4.9 < m_a < 10$ GeV range.

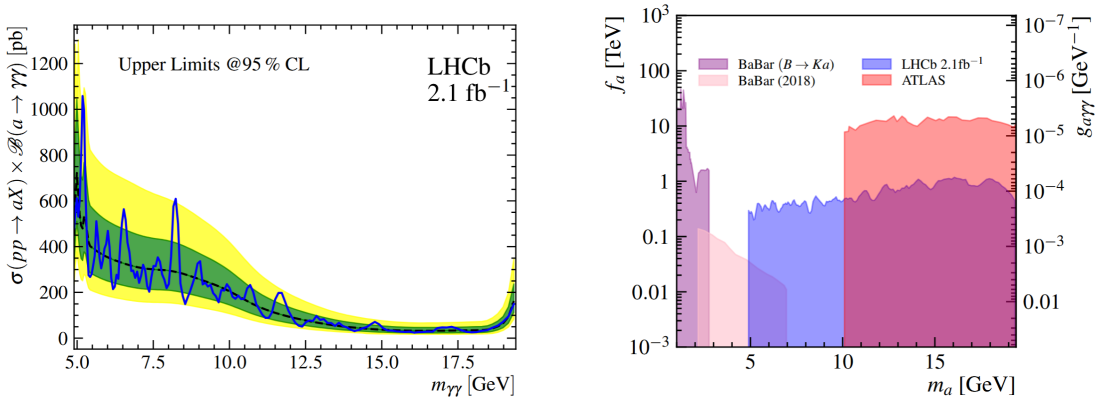


Figure 5: Left: 95% CL upper limits on $\sigma(pp \to aX) \times \mathcal{B}(a \to \gamma\gamma)$ as a function of the diphoton mass. Right: corresponding constraints on the ALP–photon coupling $g_{a\gamma\gamma}$ and symmetry-breaking scale f_a as a function of the ALP mass, compared with limits from BaBar and ATLAS. Figures from Ref. [13]

PoS(EPS-HEP2025)043

In addition to the ALP search, the same strategy and signal selection are used to investigate rare decays of heavy-flavour and bottomonium states to diphotons. Upper limits are set at 95% CL on the branching fractions $\mathcal{B}(B_s^0 \to \gamma\gamma) < 2.7 \times 10^{-5}$ and $\mathcal{B}(B^0 \to \gamma\gamma) < 0.8 \times 10^{-5}$. A first search for $\eta_b \to \gamma\gamma$ has also been performed in a hadronic environment, setting the upper limit $\sigma(pp \to \eta_b X) \times \mathcal{B}(\eta_b \to \gamma\gamma) < 765$ pb at 95% CL.

As illustrated in Figure 6, the upgraded Run 3 detector and the fully software-based trigger system will significantly enhance LHCb's reach. The trigger flexibility will enable dedicated selections for low-energy photons and long-lived pseudoscalars, extending the sensitivity to lower masses and longer lifetimes. Furthermore, searches exploiting additional decay modes, such as $a \to \pi^0\gamma$, $a \to \pi^+\pi^-\pi^0$, or $a \to \eta\gamma$, will provide access to ALPs with mixed gluon and photon couplings.

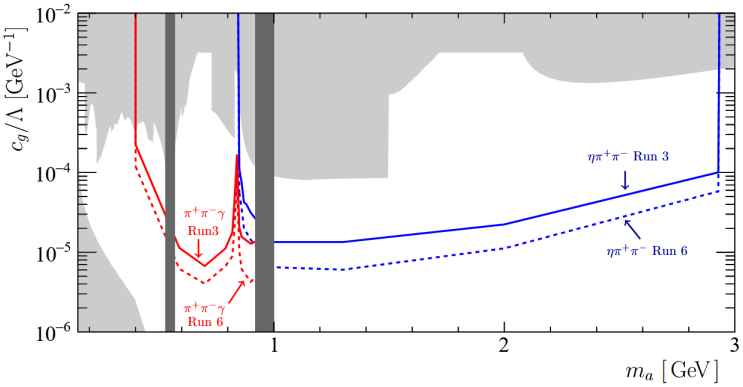


Figure 6: Projected LHCb sensitivity to axion-like particles coupled to gluons, shown in the $(m_a, c_g/\Lambda)$ plane for Runs 3 (solid) and 6 (dashed). The projections correspond to searches in the $\pi^+\pi^-\gamma$ (red) and $\eta\pi^+\pi^-$ (blue) final states. The grey regions indicate existing experimental exclusions. Figure from Ref. [7]

5. Conclusions

The LHCb experiment offers distinctive sensitivity to hidden-sector mediators that couple through the vector, scalar, and axion portals, owing to its precise vertexing, tracking, and trigger performance in the forward region. The searches reported in this proceeding probe a broad class of models predicting light and weakly interacting particles.

The dark-photon analysis, including both prompt and displaced signatures, has set competitive limits on the kinetic-mixing parameter ε over a wide mass range, from a few hundred MeV up to several tens of GeV. Searches for scalar and pseudoscalar bosons produced in $B \to K^{(*)}\chi$ decays constrain branching fractions down to $O(10^{-9})$, testing models with Higgs- or inflaton-like mixing. The recent study of axion-like particles decaying to diphotons extends the sensitivity to neutral final states and provides the first collider limits in the sub-20 GeV region, complementing results from ATLAS, CMS, and fixed-target experiments.

The ongoing Run 3 data taking, combined with the fully software-based trigger and improved reconstruction algorithms, will significantly enhance LHCb's reach. Lower trigger thresholds and refined displaced-vertex reconstruction will extend the sensitivity to longer lifetimes, smaller couplings, and a wider set of decay modes. The datasets from Runs 3 and 4 are expected to strengthen these limits, consolidating LHCb's role in exploring hidden sectors at the intensity frontier.

PoS(EPS-HEP2025)043

References

[1] Planck collaboration, N. Aghanim *et al.*, *Planck 2018 results. VI. Cosmological parameters*, *Astron. Astrophys.* **641** (2020), A6 [erratum: *Astron. Astrophys.* **652** (2021), C4] [arXiv:1807.06209 [astro-ph]].

[2] R. Essig *et al.*, *Dark Sectors and New, Light, Weakly-Coupled Particles*, Snowmass 2013 White Paper, [arXiv:1311.0029 [hep-ph]].

[3] J. Beacham *et al.*, *Physics Beyond Colliders at CERN: Beyond the Standard Model Working Group Report*, *J. Phys. G* **47** (2020), 010501 [arXiv:1901.09966 [hep-ex]].

[4] LHCb collaboration, *The LHCb Detector at the LHC*, *JINST* **3** (2008), S08005 [doi:10.1088/1748-0221/3/08/S08005], LHCb-DP-2008-001, CERN Document Server.

[5] M. Fabbrichesi, E. Gabrielli and G. Lanfranchi, *The Dark Photon*, *Eur. Phys. J. C* **80** (2020), 1098 [doi:10.1140/epjc/s10052-020-8369-3], [arXiv:2005.01515 [hep-ph]].

[6] LHCb collaboration, R. Aaij *et al.*, *Search for $A' \to \mu^+\mu^-$ decays*, *Phys. Rev. Lett.* **124** (2020), 041801 [doi:10.1103/PhysRevLett.124.041801], APS link.

[7] D. Craik, P. Ilten, D. Johnson and M. Williams, *LHCb future dark-sector sensitivity projections for Snowmass 2021*, [arXiv:2203.07048 [hep-ph]].

[8] M. Winkler, *Decay and detection of a light scalar boson mixing with the Higgs boson*, *Phys. Rev. D* **99** (2019) 015018 [arXiv:1809.01876 [hep-ph]].

[9] LHCb collaboration, R. Aaij *et al.*, *Search for hidden-sector bosons in $B^0 \to K^{*0}\mu^+\mu^-$ decays*, *Phys. Rev. Lett.* **115** (2015), 161802 [doi:10.1103/PhysRevLett.115.161802].

[10] LHCb collaboration, R. Aaij *et al.*, *Search for long-lived scalar particles in $B^+ \to K^+\chi(\mu^+\mu^-)$ decays*, *Phys. Rev. D* **95** (2017), 071101(R) [doi:10.1103/PhysRevD.95.071101].

[11] D. Craik, P. Ilten, D. Johnson and M. Williams, *LHCb future dark-sector sensitivity projections for Snowmass 2021*, [arXiv:2203.07048 [hep-ph]], [doi:10.48550/arXiv.2203.07048].

[12] H. Georgi, D. B. Kaplan and L. Randall, *Manifesting the invisible axion at low energies*, *Phys. Lett. B* **169** (1986) 73–78, [doi:10.1016/0370-2693(86)90688-X].

[13] LHCb collaboration, R. Aaij *et al.*, *Search for axion-like particles in the di-photon final state*, CERN-EP-2025-133, LHCb-PAPER-2025-012, [arXiv:2507.14390 [hep-ex]].

PoS(EPS-HEP2025)043